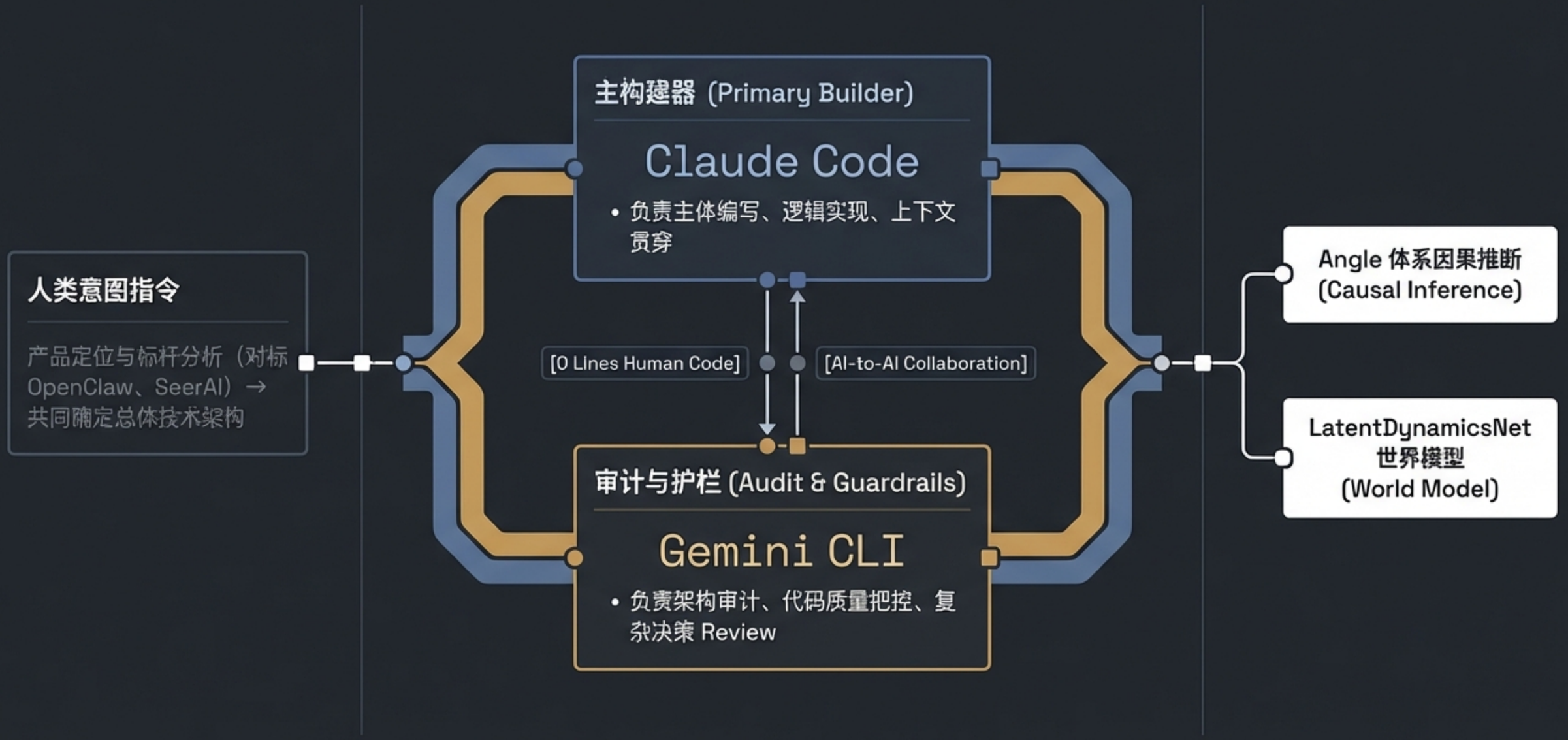
硅基造物：100% AI 驱动的软件工程革命

揭秘 GIS Data Agent v15.3 的 “零人工” 开发范式与系统结晶

[SYSTEM_READY] Agentic Dev Pipeline Engaged

NotebookLM

Agentic Dev Pipeline

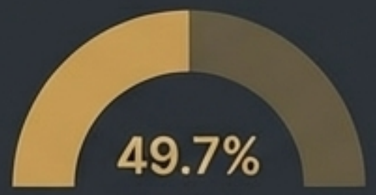


NotebookLM

118,348 行
手写等效逻辑代码

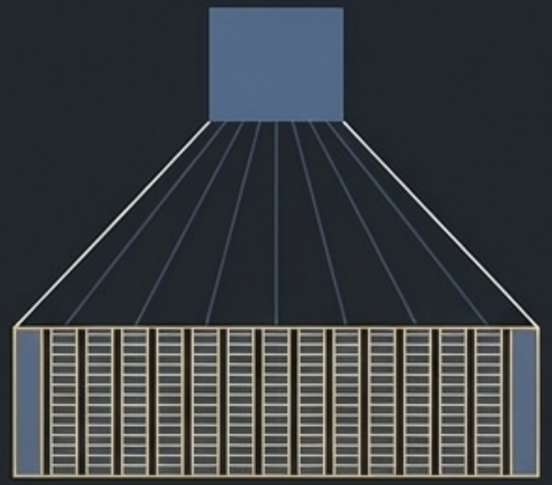
66,573 行
后端生产代码
涵盖 208 个文件，178+ 个 REST API

33,111 行
测试代码



测试覆盖率

The Documentation Funnel



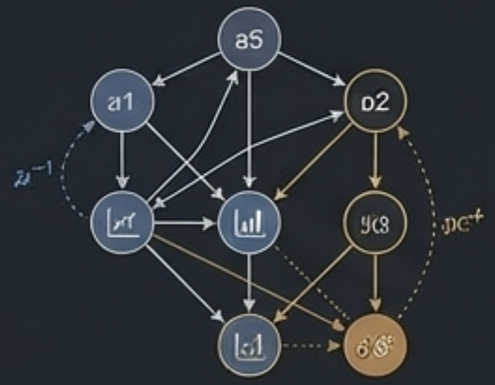
3,490 行
输入驱动类设计文档
（PRD、架构评审、因果推断方案）

总文档规模
26,824 行

21,188 行
AI 自动化产出的技术结晶
（21 章技术指南、DITA XML 结构化文档）

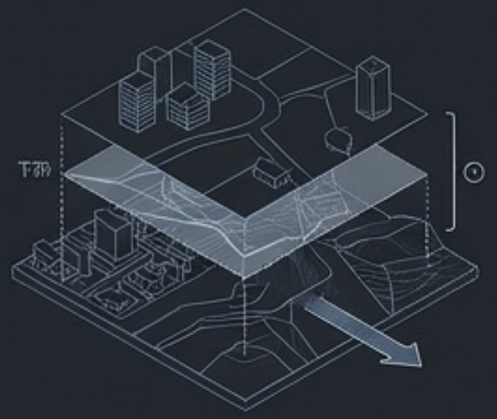
Structural

Angle A/B/C 三角度因果推理
DAG 图与反事实推演



Structural

AlphaEarth 世界模型
时空干预与溯化



Structural

3D 空间可视化
deck.gl / MapLibre 极速渲染

Structural

NL2SQL
自然语言瞬间转化为复杂空间算子

查询某区域近30天交通流量及事故相关性

↓

```
SELECT (
  SELECT
    FROM ''spatial operators',
    WHERE (
      ST_DWithin(Re.footprint,
      coalesce.properties(id,'id'),
      TIMESTAMPSESSION) TO TS, SE:defined(123),
      dataaggregated(10, 20)
  FROM
)
```

NotebookLM